Problem Set 1

Md. Fuad Al Alam

March 29, 2026

Problem 1. (Turkey Junior 2025 P3) Find all positive integers n such that the expression

$$3n^3 - 7n^2 - 15n + 35$$

is equal to the square of an integer.

Problem 2. (Korea 2023 P1) A sequence of positive reals $\{a_n\}$ is defined below

$$a_0 = 1, \quad a_1 = 3, \quad a_{n+2} = \frac{a_{n+1}^2 + 2}{a_n}$$

Show that for all non-negative integers n, a_n is a positive integer.

Problem 3. (Tuymaada 2022 Junior P5) Each row of a 24×8 table contains some permutation of the numbers $1, 2, \ldots, 8$. In each column, the numbers are multiplied. What is the minimum possible sum of all the products?

Problem 4. (Benelux 2021 P3) A cyclic quadrilateral $ABXC$ has circumcentre O. Let D be a point on line BX such that $AD = BD$. Let E be a point on line CX such that $AE = CE$. Prove that the circumcenter of $\triangle DEX$ lies on the perpendicular bisector of OA.

Problem 5. (Belarus 2025 P5) Find the smallest positive integer n such that both $n^3 - n$ and $(n+1)^3 - (n+1)$ are divisible by 2025.

Problem 6. (Germany 2025 P1) Solve the system of equations in $\mathbb{R}$:

$$\frac{a}{c} = b - \sqrt{b} + c$$
$$\sqrt{\frac{a}{c}} = \sqrt{b} + 1$$
$$\sqrt[4]{\frac{a}{c}} = \sqrt[3]{b} - 1$$

Problem 7. (Indonesia 2025 P5) Let $A_1, A_2, \ldots, A_n$ be subsets of $X = \{1, 2, \ldots, 22\}$, each with 15 elements. It is known for each $B \subseteq X$ with 10 elements, there exist $i, j \in \{1, 2, \ldots, n\}$ such that $B \subseteq A_i \cap A_j$. Prove that $n \geq 431$.

Problem 8. (Greece JBMO TST 2025 P2) Let ABC be a right triangle with $\angle A = 90^\circ$. Outside the triangle ABC, we construct an isosceles triangle ACD such that $DA = DC = BC$. Lines BC and DA intersect at point E. Prove that $DC + DE = 2CE$.